

Mathera

Complete Curriculum Map

1080 skills

7 eras · 83 domains / branches

Era	Name	World	Domains	Skills
I	Count	First Grove	7	108
II	Operate	Forge City	6	83
III	Relate	Bridge Isles	9	146
IV	Solve	Peak Range	15	199
V	Prove	Proof Palace	15	173
VI	Motion	Tide Sea	12	174
VII	Space	Star Vault	19	197
TOTAL			83	1080

Every skill below is listed Era -> Domain -> Skill. Eras I-III titles come from your design lists. Eras IV-VII follow the v2 outline (743 skills across those four eras). Era II currently includes II.1-II.6 (II.7-II.10 not yet received).

Contents by domain

Era I Count (108 skills)

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I.3	Place Value	12
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I.6	Shape & Space	14
I.7	Measure & Data	16

Era II Operate (83 skills)

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II.3	Division	15
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Era V Prove (173 skills)

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VI.6	Applications of integrals	15
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VI.8	Parametric & polar	12
VI.9	Sequences & series	11
VI.10	Power & Taylor series	16
VI.11	Statistical inference	9
VI.12	Mechanics	9

Era VII Space (197 skills)

VII.1	Vectors in R^n	12
VII.2	Linear systems	9
VII.3	Matrix algebra	12
VII.4	Vector spaces	16
VII.5	Linear transformations	9
VII.6	Determinants	9
VII.7	Eigenvalues & eigenvectors	9
VII.8	Orthogonality	12
VII.9	Symmetric matrices & SVD	8
VII.10	Functions of several variables	6
VII.11	Partial derivatives	12
VII.12	Optimization	8
VII.13	Multiple integrals	9
VII.14	Vector calculus	18
VII.15	First-order equations	7
VII.16	Second-order linear	13
VII.17	Systems of ODEs	11
VII.18	Laplace transform	9
VII.19	Fourier series	8

Era I . Count - First Grove

108 skills . 7 domains . Things begin here.

I.1 . COUNTING (16)

- 1.01 Count objects one by one
- 1.02 Number words in order
- 1.03 Count to 20
- 1.04 Count to 50
- 1.05 Count to 120
- 1.06 Read and write numerals to 20
- 1.07 How many in all
- 1.08 Zero
- 1.09 See amounts to 5 at a glance
- 1.10 See amounts to 10 at a glance
- 1.11 Count on
- 1.12 Count back
- 1.13 One more, one less
- 1.14 Skip count by 10s
- 1.15 Skip count by 5s
- 1.16 Skip count by 2s

I.2 . COMPARING & ORDERING (12)

- 2.01 More or fewer by matching
- 2.02 Equal groups
- 2.03 Number line 0-10
- 2.04 Compare numbers to 10
- 2.05 Number line 0-20
- 2.06 Order numbers
- 2.07 Ordinal numbers
- 2.08 Compare to 20
- 2.09 Compare to 100
- 2.10 Symbols =, > and <
- 2.11 Estimate a quantity
- 2.12 How many more, how many fewer

I.3 . PLACE VALUE (12)

- 3.01 Ten ones make a ten
- 3.02 Teen numbers are a ten and some ones
- 3.03 Tens and ones to 50
- 3.04 Tens and ones to 99
- 3.05 Bundle and unbundle
- 3.06 Hundreds
- 3.07 Three-digit place value
- 3.08 Expanded form
- 3.09 Number names
- 3.10 10 more, 10 less
- 3.11 The hundred chart
- 3.12 Compare using place value

I.4 . ADDITION & SUBTRACTION (23)

4.01	Put together
4.02	Take away
4.03	Addition stories
4.04	Subtraction stories
4.05	Part-part-whole
4.06	Number bonds to 5
4.07	Number bonds to 10
4.08	Doubles
4.09	Near doubles
4.10	Make ten
4.11	Count on to add
4.12	Add in any order
4.13	Subtract by counting back
4.14	Subtract by counting up
4.15	Subtract through ten
4.16	Fact families
4.17	Missing addends
4.18	The equal sign as a balance
4.19	Addition facts within 20
4.20	Subtraction facts within 20
4.21	Add and subtract tens
4.22	Two-digit addition with models
4.23	Two-digit subtraction with models

I.5 . PATTERNS & LOGIC (15)

5.01	Copy a pattern
5.02	Extend a repeating pattern
5.03	Patterns and rules
5.04	Growing patterns
5.05	Number patterns
5.06	Sort by one attribute
5.07	Sort by two attributes
5.08	Odd and even
5.09	Same and different
5.10	True or false
5.11	Always, sometimes, never
5.12	If-then
5.13	Logic grid puzzles
5.14	Order by clues
5.15	Guess my number

I.6 . SHAPE & SPACE (14)

6.01	Name flat shapes
6.02	Shape features
6.03	Shapes in any position
6.04	Name solid shapes
6.05	Solid shape features
6.06	Put shapes together
6.07	Take shapes apart
6.08	Solids from blocks

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- | | |
|------|----------------------|
| 6.09 | Position words |
| 6.10 | Directions on a grid |
| 6.11 | Halves |
| 6.12 | Quarters |
| 6.13 | Symmetry |
| 6.14 | Shape riddles |

I.7 . MEASURE & DATA (16)

- | | |
|------|-----------------------|
| 7.01 | Compare lengths |
| 7.02 | Order by length |
| 7.03 | Nonstandard units |
| 7.04 | Measure with a ruler |
| 7.05 | Compare weights |
| 7.06 | Compare capacity |
| 7.07 | Time to the hour |
| 7.08 | Time to the half hour |
| 7.09 | Days and months |
| 7.10 | Read a calendar |
| 7.11 | Coins |
| 7.12 | Make amounts |
| 7.13 | Tally marks |
| 7.14 | Picture graphs |
| 7.15 | Bar graphs |
| 7.16 | Ask with data |

Era II . Operate - Forge City

83 skills . 6 domains . Do things to numbers.

II.1 . PLACE VALUE TO MILLIONS (12)

- 1.01 Thousands
- 1.02 Numbers to a million
- 1.03 Place value patterns
- 1.04 Compare and order large numbers
- 1.05 Round to tens and hundreds
- 1.06 Round large numbers
- 1.07 Estimate sums and differences
- 1.08 Column addition, no regrouping
- 1.09 Column addition with regrouping
- 1.10 Column subtraction, no regrouping
- 1.11 Column subtraction with regrouping
- 1.12 Addition and subtraction problems

II.2 . MULTIPLICATION (15)

- 2.01 Equal groups
- 2.02 Arrays
- 2.03 Multiply by 0, 1, 2, 5 and 10
- 2.04 Facts for 3, 4 and 6
- 2.05 Facts for 7, 8 and 9
- 2.06 All facts to 10 x 10
- 2.07 The area model
- 2.08 Properties of multiplication
- 2.09 Multiply by multiples of 10
- 2.10 Two-digit by one-digit
- 2.11 Three- and four-digit by one-digit
- 2.12 Two-digit by two-digit
- 2.13 Multi-digit multiplication
- 2.14 Multiplication problems
- 2.15 Multiplication patterns

II.3 . DIVISION (15)

- 3.01 Sharing equally
- 3.02 Making groups
- 3.03 Division as a missing factor
- 3.04 Division facts
- 3.05 Dividing with 0 and 1
- 3.06 Remainders
- 3.07 Interpreting remainders
- 3.08 Divide multiples of 10
- 3.09 Two-digit by one-digit division
- 3.10 Long division by one digit
- 3.11 Division by two digits
- 3.12 Divisibility rules
- 3.13 Division problems
- 3.14 Mixed operations
- 3.15 Estimating quotients

II.4 . FACTORS & MULTIPLES (12)

4.01	Factor pairs
4.02	Multiples
4.03	Primes and composites
4.04	The sieve of Eratosthenes
4.05	Prime factorization
4.06	Common factors
4.07	Using the GCF
4.08	Common multiples
4.09	Using the LCM
4.10	Square numbers
4.11	Even and odd rules
4.12	Number puzzles with factors

II.5 . FRACTIONS I (15)

5.01	Equal parts
5.02	Unit fractions
5.03	Fractions of a whole
5.04	Fractions of a set
5.05	Fractions on a number line
5.06	Whole numbers as fractions
5.07	Equivalent fractions with models
5.08	Equivalent fractions by rule
5.09	Compare, same denominator
5.10	Compare, same numerator
5.11	Compare any fractions
5.12	Mixed numbers
5.13	Improper fractions
5.14	Fractions as division
5.15	Fraction sense

II.6 . FRACTIONS II (14)

6.01	Add, like denominators
6.02	Subtract, like denominators
6.03	Break fractions apart
6.04	Mixed numbers, like denominators
6.05	Common denominators
6.06	Add, unlike denominators
6.07	Subtract, unlike denominators
6.08	Mixed numbers, unlike denominators
6.09	Fraction x whole number
6.10	Multiply fractions
6.11	Multiplying as scaling
6.12	Unit fraction / whole number
6.13	Whole number / unit fraction
6.14	Fraction problems

Era III . Relate - Bridge Isles

146 skills . 9 domains . How quantities connect.

III.1 . INTEGERS & RATIONALS (15)

- 1.01 Negative numbers in context
- 1.02 Integers on a number line
- 1.03 Absolute value
- 1.04 Add integers
- 1.05 Subtract integers
- 1.06 Multiply integers
- 1.07 Divide integers
- 1.08 Order of operations with integers
- 1.09 Rational numbers
- 1.10 Add and subtract rationals
- 1.11 Multiply and divide rationals
- 1.12 Terminating and repeating decimals
- 1.13 Repeating decimals to fractions
- 1.14 Compare and order rationals
- 1.15 All four quadrants

III.2 . RATIOS & RATES (15)

- 2.01 Ratio language
- 2.02 Equivalent ratios
- 2.03 Ratio tables
- 2.04 Unit rates
- 2.05 Rates with fractions
- 2.06 Ratios on a graph
- 2.07 Proportions
- 2.08 Proportional relationships
- 2.09 The constant of proportionality
- 2.10 Scale drawings
- 2.11 Unit conversion with ratios
- 2.12 Sharing in a ratio
- 2.13 Speed, distance, time
- 2.14 Comparing ratios
- 2.15 Ratio puzzles

III.3 . PERCENT (17)

- 3.01 Percent means per hundred
- 3.02 Fractions, decimals, percents
- 3.03 Percent of a number
- 3.04 Find the whole
- 3.05 Find the percent
- 3.06 Over 100% and under 1%
- 3.07 Percent increase
- 3.08 Percent decrease
- 3.09 Discounts and sale prices
- 3.10 Tax and tips
- 3.11 Markup and profit
- 3.12 Simple interest

- 3.13 Percent error
- 3.14 Percent change traps
- 3.15 Percent in data
- 3.16 Estimating with percents
- 3.17 Multi-step percent problems

III.4 . EXPONENTS & ROOTS (17)

- 4.01 Whole-number exponents
- 4.02 Powers of ten
- 4.03 Exponents in the order of operations
- 4.04 Multiplying powers
- 4.05 Dividing powers
- 4.06 Power of a power
- 4.07 Negative exponents
- 4.08 Scientific notation
- 4.09 Calculating in scientific notation
- 4.10 Comparing huge and tiny numbers
- 4.11 Square roots
- 4.12 Cube roots
- 4.13 Estimate roots
- 4.14 Irrational numbers
- 4.15 The real number system
- 4.16 Compare irrationals
- 4.17 Growth with exponents

III.5 . EXPRESSIONS (15)

- 5.01 Variables
- 5.02 Expressions from words
- 5.03 Evaluate expressions
- 5.04 Parts of an expression
- 5.05 Like terms
- 5.06 The distributive property
- 5.07 Factor out a GCF
- 5.08 Equivalent expressions
- 5.09 Simplify expressions
- 5.10 Expressions in geometry
- 5.11 Expressions in context
- 5.12 Expressions with powers
- 5.13 Patterns to expressions
- 5.14 Algebra tiles
- 5.15 Why number tricks work

III.6 . EQUATIONS & INEQUALITIES (12)

- 6.01 What an equation is
- 6.02 One-step equations: + and -
- 6.03 One-step equations: x and /
- 6.04 Two-step equations
- 6.05 Multi-step equations
- 6.06 Variables on both sides
- 6.07 Equations from word problems
- 6.08 Inequalities

- 6.09 One-step inequalities
- 6.10 Two-step inequalities
- 6.11 Rearrange formulas
- 6.12 Choosing a strategy

III.7 . LINES & FIRST FUNCTIONS (12)

- 7.01 Input-output rules
- 7.02 Graph from a table
- 7.03 Rate of change
- 7.04 Slope
- 7.05 Slope and similar triangles
- 7.06 $y = mx + b$
- 7.07 Proportional vs non-proportional
- 7.08 Linear vs nonlinear
- 7.09 Functions
- 7.10 Compare functions
- 7.11 Describe graphs qualitatively
- 7.12 Linear models

III.8 . GEOMETRY (23)

- 8.01 Angle relationships
- 8.02 Parallel lines & a transversal
- 8.03 Triangle angle sum
- 8.04 Area of parallelograms
- 8.05 Area of triangles
- 8.06 Area of trapezoids
- 8.07 Composite area
- 8.08 Circle parts
- 8.09 Circumference
- 8.10 Area of a circle
- 8.11 Surface area of prisms
- 8.12 Volume of prisms
- 8.13 Volume of cylinders
- 8.14 Volume of cones & spheres
- 8.15 Pythagorean theorem
- 8.16 Converse of Pythagoras
- 8.17 Distance on the plane
- 8.18 Pythagoras in 3D
- 8.19 Translations
- 8.20 Reflections
- 8.21 Rotations
- 8.22 Dilations
- 8.23 Congruence & similarity informally

III.9 . STATISTICS & PROBABILITY (20)

- 9.01 Statistical questions
- 9.02 Mean
- 9.03 Median & mode
- 9.04 Range
- 9.05 Quartiles & IQR
- 9.06 Box plots

9.07	Dot plots
9.08	Histograms
9.09	Mean absolute deviation
9.10	Sampling
9.11	Inferences from samples
9.12	Compare populations
9.13	Probability basics
9.14	Experimental probability
9.15	Sample spaces
9.16	Compound events
9.17	Simulations
9.18	Scatter plots
9.19	Trend lines
9.20	Two-way tables

Era IV . Solve - Peak Range

199 skills . 15 domains . Find the unknown.

IV.1 . EQUATIONS & INEQUALITIES (5)

- 1.01 Solve one-step equations
- 1.02 Solve two-step equations
- 1.03 Solve multi-step equations
- 1.04 Solve and graph inequalities
- 1.05 Compound inequalities

IV.2 . FUNCTIONS (15)

- 2.01 Function or not
- 2.02 Function notation $f(x)$
- 2.03 Evaluate functions
- 2.04 Domain from a graph
- 2.05 Range from a graph
- 2.06 Domain from an equation
- 2.07 Vertical line test
- 2.08 Independent & dependent
- 2.09 Discrete vs continuous
- 2.10 Increasing & decreasing
- 2.11 Maxima & minima (visual)
- 2.12 Average rate of change
- 2.13 Piecewise graphs
- 2.14 Even & odd (visual)
- 2.15 Inverse from a table

IV.3 . LINEAR FUNCTIONS (9)

- 3.01 Slope from two points
- 3.02 Slope from a graph
- 3.03 Slope-intercept form
- 3.04 Point-slope form
- 3.05 Standard form
- 3.06 Graph from intercepts
- 3.07 Parallel & perpendicular slopes
- 3.08 Write equation from graph
- 3.09 Model with linear functions

IV.4 . SYSTEMS (15)

- 4.01 Solve by graphing
- 4.02 Solve by substitution
- 4.03 Solve by elimination
- 4.04 Systems of inequalities
- 4.05 Three-variable intro
- 4.06 Consistent & inconsistent
- 4.07 Dependent systems
- 4.08 Word: mixture
- 4.09 Word: motion
- 4.10 Word: value
- 4.11 Break-even models

- 4.12 Linear programming region
- 4.13 Matrix row ops intro
- 4.14 Augmented matrices
- 4.15 Gaussian elimination (2x2)

IV.5 . EXPONENTS & RADICALS (7)

- 5.01 Product & quotient rules
- 5.02 Power rules
- 5.03 Negative exponents
- 5.04 Rational exponents
- 5.05 Simplify radicals
- 5.06 Operations with radicals
- 5.07 Rationalize denominators

IV.6 . POLYNOMIALS (16)

- 6.01 Classify by degree & terms
- 6.02 Add & subtract polynomials
- 6.03 Multiply monomials
- 6.04 FOIL & distribute
- 6.05 Special products
- 6.06 GCF factoring
- 6.07 Factor trinomials $a=1$
- 6.08 Factor trinomials $a \neq 1$
- 6.09 Difference of squares
- 6.10 Sum/diff of cubes
- 6.11 Factor by grouping
- 6.12 Factor completely
- 6.13 Divide by monomial
- 6.14 Long division of polynomials
- 6.15 Synthetic division
- 6.16 Remainder theorem

IV.7 . QUADRATICS (17)

- 7.01 Graph parabolas from vertex
- 7.02 Axis of symmetry
- 7.03 Intercept form
- 7.04 Vertex form conversions
- 7.05 Complete the square
- 7.06 Quadratic formula
- 7.07 Discriminant
- 7.08 Complex roots intro
- 7.09 Projectile models
- 7.10 Max/min word problems
- 7.11 Factor to solve
- 7.12 Square-root method
- 7.13 Systems: linear-quadratic
- 7.14 Quadratic inequalities
- 7.15 Transformations of $y=x^2$
- 7.16 Focus-directrix intro
- 7.17 Fit a quadratic

IV.8 . COMPLEX NUMBERS (16)

8.01	Imaginary unit i
8.02	Powers of i
8.03	Add & subtract complex
8.04	Multiply complex
8.05	Complex conjugates
8.06	Divide complex
8.07	Modulus
8.08	Argand plane
8.09	Polar form
8.10	De Moivre (powers)
8.11	n th roots of complex
8.12	Complex from quadratics
8.13	Operations in polar form
8.14	Euler's formula glimpse
8.15	Complex conjugates & graphs
8.16	Applications of complex

IV.9 . FUNCTION TOOLKIT (18)

9.01	Parent functions gallery
9.02	Vertical shifts
9.03	Horizontal shifts
9.04	Reflections
9.05	Stretches & compressions
9.06	Compose transformations
9.07	$f(g(x))$ composition
9.08	Decompose functions
9.09	One-to-one & inverses
9.10	Find inverse algebraically
9.11	Graph inverse
9.12	Restrict domain for inverse
9.13	Absolute value graphs
9.14	Step & piecewise build
9.15	Combine functions (+ - $\cdot$ /)
9.16	Difference quotient
9.17	Symmetry tests
9.18	End behavior (visual)

IV.10 . POLYNOMIAL FUNCTIONS (14)

10.01	End behavior & degree
10.02	Zeros & multiplicity
10.03	Sketch from factors
10.04	Intermediate Value Theorem
10.05	Rational Root Theorem
10.06	Factor theorem applications
10.07	Complex zeros conjugate pairs
10.08	Descartes' rule of signs
10.09	Upper/lower bounds
10.10	Graph with calculus-free tools
10.11	Polynomial inequalities
10.12	Model with polynomials

- 10.13 Compare polynomial models
- 10.14 Turning points bound

IV.11 . RATIONAL & RADICAL FUNCTIONS (14)

- 11.01 Simplify rational expressions
- 11.02 Multiply/divide rationals
- 11.03 Add/subtract rationals
- 11.04 Complex fractions
- 11.05 Vertical asymptotes
- 11.06 Horizontal & slant asymptotes
- 11.07 Holes in graphs
- 11.08 Graph rational functions
- 11.09 Solve rational equations
- 11.10 Rational inequalities
- 11.11 Square-root functions
- 11.12 Cube-root functions
- 11.13 Solve radical equations
- 11.14 Extraneous roots

IV.12 . EXPONENTIAL & LOGARITHMIC (17)

- 12.01 Exponential growth graphs
- 12.02 Exponential decay
- 12.03 e and continuous growth
- 12.04 Log as inverse of exp
- 12.05 Common & natural logs
- 12.06 Log properties: product
- 12.07 Log properties: quotient/power
- 12.08 Change of base
- 12.09 Solve exponential equations
- 12.10 Solve log equations
- 12.11 Exponential models (finance)
- 12.12 Half-life & doubling
- 12.13 Newton's law of cooling
- 12.14 Logarithmic scales
- 12.15 Graph log functions
- 12.16 Transform exp & log
- 12.17 Compare growth rates

IV.13 . SEQUENCES & SERIES (14)

- 13.01 Sequence notation a_n
- 13.02 Arithmetic sequences
- 13.03 Arithmetic series
- 13.04 Geometric sequences
- 13.05 Geometric series (finite)
- 13.06 Infinite geometric $|r| < 1$
- 13.07 Recursive definitions
- 13.08 Explicit from recursive
- 13.09 Sigma notation
- 13.10 Partial sums
- 13.11 Fibonacci & variants
- 13.12 Applications of series

13.13 Binomial coefficients

13.14 Binomial theorem

IV.14 . COUNTING & PROBABILITY (12)

14.01 Fundamental counting

14.02 Permutations

14.03 Combinations

14.04 Distinguishable permutations

14.05 Pascal & combinations

14.06 Sample spaces

14.07 Theoretical probability

14.08 Independent events

14.09 Dependent events

14.10 Mutually exclusive

14.11 Conditional probability

14.12 Expected value intro

IV.15 . DATA & DISTRIBUTIONS (10)

15.01 Dot plots & histograms

15.02 Mean, median, mode

15.03 Range & IQR

15.04 Box plots

15.05 Variance & SD

15.06 Z-scores

15.07 Normal curve intro

15.08 Empirical rule

15.09 Scatter & correlation

15.10 Least-squares line

Era V . Prove - Proof Palace

173 skills . 15 domains . Know it's true.

V.1 . LOGIC & PROOF (16)

- 1.01 Statements & negations
- 1.02 Conditional statements
- 1.03 Converse, inverse, contrapositive
- 1.04 Biconditionals
- 1.05 Inductive reasoning
- 1.06 Deductive reasoning
- 1.07 Laws of detachment & syllogism
- 1.08 Algebraic proof
- 1.09 Two-column proof intro
- 1.10 Paragraph proofs
- 1.11 Indirect proof (contradiction)
- 1.12 Counterexamples
- 1.13 Quantifiers ? ?
- 1.14 Set language in proofs
- 1.15 Proof planning
- 1.16 Common proof patterns

V.2 . LINES & ANGLES (6)

- 2.01 Angle pairs & complements
- 2.02 Vertical angles
- 2.03 Parallel lines & transversals
- 2.04 Prove lines parallel
- 2.05 Angle sum on a line
- 2.06 Perpendicular theorems

V.3 . TRIANGLES (14)

- 3.01 Triangle classification
- 3.02 Triangle angle sum
- 3.03 Exterior angle theorem
- 3.04 SSS congruence
- 3.05 SAS congruence
- 3.06 ASA & AAS
- 3.07 HL for right triangles
- 3.08 CPCTC
- 3.09 Isosceles triangle theorems
- 3.10 Triangle inequality
- 3.11 Hinge theorem
- 3.12 Medians & centroid
- 3.13 Altitude & orthocenter
- 3.14 Perp. bisectors & circumcenter

V.4 . POLYGONS (9)

- 4.01 Interior angle sum
- 4.02 Exterior angles of polygons
- 4.03 Regular polygons
- 4.04 Parallelogram properties

- 4.05 Rectangle, rhombus, square
- 4.06 Trapezoids & kites
- 4.07 Prove parallelogram
- 4.08 Coordinate proof of quads
- 4.09 Area formulas review

V.5 . SIMILARITY (12)

- 5.01 Similar figures definition
- 5.02 AA similarity
- 5.03 SSS & SAS similarity
- 5.04 Corresponding lengths
- 5.05 Triangle proportionality
- 5.06 Angle bisector theorem
- 5.07 Similar right triangles
- 5.08 Geometric mean in right ?
- 5.09 Scale drawings
- 5.10 Indirect measurement
- 5.11 Areas of similar figures
- 5.12 Volumes of similar solids

V.6 . RIGHT-TRIANGLE TRIG (14)

- 6.01 Pythagorean theorem
- 6.02 Converse of Pythagoras
- 6.03 Special 45-45-90
- 6.04 Special 30-60-90
- 6.05 Sine & cosine ratios
- 6.06 Tangent ratio
- 6.07 Solve right triangles
- 6.08 Angle of elevation/depression
- 6.09 Inverse trig for angles
- 6.10 Cofunction identities
- 6.11 Pythagorean identity intro
- 6.12 Applications: navigation
- 6.13 Applications: surveying
- 6.14 Trig in coordinate plane

V.7 . GENERAL TRIANGLES (9)

- 7.01 Law of sines
- 7.02 Ambiguous case SSA
- 7.03 Law of cosines
- 7.04 Area: $(1/2)ab \sin C$
- 7.05 Heron's formula
- 7.06 Solve SAS / ASA / SSA
- 7.07 Solve SSS / SAS
- 7.08 Bearing problems
- 7.09 Vectors in the plane intro

V.8 . CIRCLES (17)

- 8.01 Circle vocabulary
- 8.02 Central angles & arcs
- 8.03 Inscribed angles

8.04	Angles from chords
8.05	Tangents from a point
8.06	Tangent-chord angles
8.07	Intersecting chords
8.08	Secant & tangent lengths
8.09	Arc length
8.10	Sector area
8.11	Segment area
8.12	Equation of a circle
8.13	Complete square for circle
8.14	Inscribed & circumscribed
8.15	Cyclic quadrilaterals
8.16	Power of a point
8.17	Common tangent constructions

V.9 . TRIG FUNCTIONS (15)

9.01	Unit circle definition
9.02	Radians & degrees
9.03	Coterminal angles
9.04	Reference angles
9.05	Sine & cosine graphs
9.06	Amplitude & period
9.07	Phase shift
9.08	Tangent graph
9.09	Sec, csc, cot graphs
9.10	Domain & range of trig
9.11	Inverse sine/cosine/tan
9.12	Solve for angles on circle
9.13	Sinusoidal models
9.14	Harmonic motion intro
9.15	Polar angle sense

V.10 . IDENTITIES & EQUATIONS (12)

10.01	Reciprocal identities
10.02	Quotient identities
10.03	Pythagorean identities
10.04	Even/odd trig
10.05	Cofunction identities
10.06	Sum & difference formulas
10.07	Double-angle formulas
10.08	Half-angle formulas
10.09	Verify identities
10.10	Solve trig equations (basic)
10.11	Solve trig equations (multi)
10.12	Product-to-sum glimpse

V.11 . COORDINATE GEOMETRY (9)

11.01	Distance & midpoint
11.02	Partition a segment
11.03	Slope criteria proofs
11.04	Equation of a line review

- 11.05 Parallel/perp in coordinates
- 11.06 Coordinate triangle proofs
- 11.07 Locus problems
- 11.08 Area via coordinates
- 11.09 Transformations on plane

V.12 . CONIC SECTIONS (11)

- 12.01 Parabola focus-directrix
- 12.02 Parabola equations
- 12.03 Ellipse definition
- 12.04 Ellipse standard form
- 12.05 Hyperbola definition
- 12.06 Hyperbola standard form
- 12.07 Classify conics
- 12.08 Complete the square (conics)
- 12.09 Translate conics
- 12.10 Eccentricity
- 12.11 Applications of conics

V.13 . TRANSFORMATIONS (12)

- 13.01 Translations
- 13.02 Reflections
- 13.03 Rotations 90/180/270
- 13.04 Dilations
- 13.05 Composition of isometries
- 13.06 Glide reflections
- 13.07 Symmetry groups intro
- 13.08 Mapping figures
- 13.09 Coordinate rules
- 13.10 Invariant properties
- 13.11 Tessellations
- 13.12 Frieze patterns glimpse

V.14 . SOLIDS (8)

- 14.01 Prisms & cylinders
- 14.02 Pyramids & cones
- 14.03 Spheres
- 14.04 Surface area
- 14.05 Volume formulas
- 14.06 Cavalieri's principle
- 14.07 Cross sections
- 14.08 Nets & 3D reasoning

V.15 . CONSTRUCTIONS (9)

- 15.01 Copy a segment
- 15.02 Copy an angle
- 15.03 Perpendicular bisector
- 15.04 Angle bisector
- 15.05 Parallel through a point
- 15.06 Equilateral triangle
- 15.07 Incircle & circumcircle

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- 15.08 Tangent from external point
 - 15.09 Classical impossible glimpse

Era VI . Motion - Tide Sea

174 skills . 12 domains . Understand change.

VI.1 . LIMITS & CONTINUITY (20)

1.01	Intuitive limit idea
1.02	Limit from a table
1.03	Limit from a graph
1.04	One-sided limits
1.05	Limit laws
1.06	Direct substitution
1.07	Factoring to find limits
1.08	Rationalize for limits
1.09	Limits at infinity
1.10	Infinite limits & asymptotes
1.11	Squeeze theorem
1.12	Trig limits $(\sin x)/x$
1.13	Continuity at a point
1.14	Types of discontinuity
1.15	IVT applications
1.16	Limits of piecewise
1.17	ϵ - δ definition intro
1.18	Limit of a sequence
1.19	Unbounded behavior
1.20	Comparing growth (limits)

VI.2 . DERIVATIVES (18)

2.01	Tangent slope idea
2.02	Definition of derivative
2.03	Derivative as function
2.04	Differentiability vs continuity
2.05	Power rule
2.06	Constant & sum rules
2.07	Product rule
2.08	Quotient rule
2.09	Chain rule
2.10	Derivatives of trig
2.11	Derivatives of exp & log
2.12	Implicit differentiation
2.13	Higher-order derivatives
2.14	Derivative from graph
2.15	Numerical derivatives
2.16	Logarithmic differentiation
2.17	Inverse function derivatives
2.18	Related rates setup

VI.3 . APPLICATIONS OF DERIVATIVES (21)

3.01	Position, velocity, accel
3.02	Related rates problems
3.03	Linearization
3.04	Differentials

3.05	Extreme Value Theorem
3.06	Critical points
3.07	Closed interval method
3.08	Mean Value Theorem
3.09	Rolle's theorem
3.10	First derivative test
3.11	Second derivative test
3.12	Concavity & inflection
3.13	Curve sketching
3.14	Optimization (geometry)
3.15	Optimization (business)
3.16	Newton's method
3.17	L'Hôpital's rule
3.18	Indeterminate forms
3.19	Motion along a line
3.20	Marginal analysis
3.21	Summary: f , f' , f''

VI.4 . INTEGRALS (15)

4.01	Area under a curve idea
4.02	Riemann sums
4.03	Definite integral definition
4.04	Midpoint & trapezoid rules
4.05	Antiderivatives
4.06	Indefinite integrals
4.07	Fundamental Theorem I
4.08	Fundamental Theorem II
4.09	Net change theorem
4.10	Substitution (u-sub)
4.11	Integrals of trig
4.12	Integrals of exp & log
4.13	Even & odd integrands
4.14	Average value of a function
4.15	Accumulation functions

VI.5 . INTEGRATION TECHNIQUES (16)

5.01	Integration by parts
5.02	Trig integrals (powers)
5.03	Trig substitution
5.04	Partial fractions (linear)
5.05	Partial fractions (quadratic)
5.06	Improper integrals (inf)
5.07	Improper integrals (discont.)
5.08	Completing square in integrals
5.09	Rationalizing substitutions
5.10	Tables of integrals
5.11	Numerical integration error
5.12	Strategy: which method?
5.13	Reduction formulas glimpse
5.14	Integrals yielding inverse trig
5.15	Hyperbolic intro (optional)

5.16 CAS vs by-hand judgment

VI.6 . APPLICATIONS OF INTEGRALS (15)

- 6.01 Area between curves
- 6.02 Disk method
- 6.03 Washer method
- 6.04 Shell method
- 6.05 Arc length
- 6.06 Surface of revolution
- 6.07 Work (springs, pumps)
- 6.08 Hydrostatic force
- 6.09 Moments & center of mass
- 6.10 Probability density intro
- 6.11 Present & future value
- 6.12 Biology: cardiac output
- 6.13 Consumer/producer surplus
- 6.14 Volume by known cross-section
- 6.15 Compare volume methods

VI.7 . DIFFERENTIAL EQUATIONS I (12)

- 7.01 What is a DE?
- 7.02 Solution verification
- 7.03 Direction fields
- 7.04 Euler's method
- 7.05 Separable equations
- 7.06 Exponential growth/decay DE
- 7.07 Newton's law of cooling DE
- 7.08 Logistic equation
- 7.09 Mixing problems
- 7.10 First-order linear (?)
- 7.11 Existence & uniqueness idea
- 7.12 Autonomous equations

VI.8 . PARAMETRIC & POLAR (12)

- 8.01 Parametric equations
- 8.02 Eliminate the parameter
- 8.03 Derivatives dy/dx parametric
- 8.04 Arc length parametric
- 8.05 Polar coordinates
- 8.06 Convert polar ? cartesian
- 8.07 Graph polar roses & limaçons
- 8.08 Tangents in polar
- 8.09 Area in polar
- 8.10 Arc length in polar
- 8.11 Conics in polar
- 8.12 Parametric motion apps

VI.9 . SEQUENCES & SERIES (11)

- 9.01 Sequence convergence
- 9.02 Monotone & bounded
- 9.03 Series & partial sums

9.04	Geometric series review
9.05	Telescoping series
9.06	Divergence test
9.07	Integral test
9.08	Comparison tests
9.09	Alternating series
9.10	Ratio & root tests
9.11	Absolute vs conditional

VI.10 . POWER & TAYLOR SERIES (16)

10.01	Power series idea
10.02	Radius of convergence
10.03	Interval of convergence
10.04	Differentiate power series
10.05	Integrate power series
10.06	Taylor polynomials
10.07	Taylor & Maclaurin series
10.08	Common Maclaurin series
10.09	Binomial series
10.10	Remainder estimates
10.11	Approximate integrals
10.12	Solving DE with series
10.13	Analytic functions glimpse
10.14	Fourier vs Taylor contrast
10.15	Applications: constants
10.16	Power series manipulations

VI.11 . STATISTICAL INFERENCE (9)

11.01	Sampling distributions
11.02	CLT idea
11.03	Confidence intervals (mean)
11.04	Confidence intervals (prop.)
11.05	Hypothesis test logic
11.06	p-values
11.07	t-procedures
11.08	Chi-square intro
11.09	Type I / Type II errors

VI.12 . MECHANICS (9)

12.01	Displacement from velocity
12.02	Work-energy glimpse
12.03	Hooke's law & springs
12.04	Center of mass (calc)
12.05	Projectile via calc
12.06	Simple harmonic from DE
12.07	Escape velocity idea
12.08	Kepler glimpse
12.09	Force & acceleration models

Era VII . Space - Star Vault

197 skills . 19 domains . Explore higher worlds.

VII.1 . VECTORS IN R^n (12)

- 1.01 Vector geometry in R^2/R^3
- 1.02 Vector addition & scaling
- 1.03 Dot product
- 1.04 Orthogonality
- 1.05 Projections
- 1.06 Cross product
- 1.07 Lines in R^3
- 1.08 Planes in R^3
- 1.09 Distances: point-line-plane
- 1.10 Linear combinations
- 1.11 Span of vectors
- 1.12 Norm & unit vectors

VII.2 . LINEAR SYSTEMS (9)

- 2.01 Row echelon form
- 2.02 Reduced row echelon
- 2.03 Gaussian elimination
- 2.04 Free variables & params
- 2.05 Homogeneous systems
- 2.06 Consistency criteria
- 2.07 Rank idea
- 2.08 Applications: networks
- 2.09 Applications: balance

VII.3 . MATRIX ALGEBRA (12)

- 3.01 Matrix operations
- 3.02 Matrix multiplication
- 3.03 Identity & inverse
- 3.04 Elementary matrices
- 3.05 Solve via inverse
- 3.06 Transpose
- 3.07 Block matrices
- 3.08 Matrix powers
- 3.09 LU factorization intro
- 3.10 Sparse vs dense
- 3.11 Computer arithmetic caution
- 3.12 Affine maps as matrices

VII.4 . VECTOR SPACES (16)

- 4.01 Axioms of a vector space
- 4.02 Subspaces
- 4.03 Null space
- 4.04 Column & row space
- 4.05 Basis
- 4.06 Dimension
- 4.07 Coordinates in a basis

- 4.08 Change of basis
- 4.09 Rank-nullity theorem
- 4.10 Examples: P_n , $C[a,b]$
- 4.11 Affine subspaces
- 4.12 Direct sums
- 4.13 Isomorphism of spaces
- 4.14 Standard bases
- 4.15 Extending independent sets
- 4.16 Extracting bases from spans

VII.5 . LINEAR TRANSFORMATIONS (9)

- 5.01 Definition & examples
- 5.02 Matrix of a linear map
- 5.03 Kernel & image
- 5.04 Composition of maps
- 5.05 Invertible transformations
- 5.06 Similarity of matrices
- 5.07 Projections as maps
- 5.08 Rotations & reflections
- 5.09 Rank of a transformation

VII.6 . DETERMINANTS (9)

- 6.01 2x2 and 3x3 determinants
- 6.02 Cofactor expansion
- 6.03 Properties of det
- 6.04 Det & invertibility
- 6.05 Det as volume
- 6.06 Cramer's rule
- 6.07 Adjugate & inverse
- 6.08 Det under row ops
- 6.09 Orientation & sign

VII.7 . EIGENVALUES & EIGENVECTORS (9)

- 7.01 Eigenvalue equation
- 7.02 Characteristic polynomial
- 7.03 Eigenspaces
- 7.04 Diagonalization
- 7.05 When diagonalizable?
- 7.06 Powers via diagonalization
- 7.07 Markov chains intro
- 7.08 Differential systems preview
- 7.09 Complex eigenvalues

VII.8 . ORTHOGONALITY (12)

- 8.01 Orthogonal sets
- 8.02 Orthonormal bases
- 8.03 Gram-Schmidt
- 8.04 QR factorization
- 8.05 Orthogonal projections
- 8.06 Least squares
- 8.07 Normal equations

- 8.08 Orthogonal matrices
- 8.09 Fourier via ortho glimpse
- 8.10 Cauchy-Schwarz
- 8.11 Triangle inequality (norms)
- 8.12 Best approximation

VII.9 . SYMMETRIC MATRICES & SVD (8)

- 9.01 Spectral theorem (sym)
- 9.02 Orthogonal diagonalization
- 9.03 Quadratic forms
- 9.04 Positive definite
- 9.05 SVD idea
- 9.06 Singular values
- 9.07 Low-rank approximation
- 9.08 PCA glimpse

VII.10 . FUNCTIONS OF SEVERAL VARIABLES (6)

- 10.01 Graphs & level curves
- 10.02 Level surfaces
- 10.03 Limits in $\mathbb{R}^2$
- 10.04 Continuity in $\mathbb{R}^2$
- 10.05 Domain in several variables
- 10.06 Quadric surfaces review

VII.11 . PARTIAL DERIVATIVES (12)

- 11.01 Partial derivatives
- 11.02 Higher partials & Clairaut
- 11.03 Tangent planes
- 11.04 Linear approximation (multi)
- 11.05 Chain rule (multi)
- 11.06 Directional derivatives
- 11.07 Gradient vector
- 11.08 Gradient & level sets
- 11.09 Implicit differentiation (multi)
- 11.10 Differentiability in $\mathbb{R}^n$
- 11.11 Jacobian matrix
- 11.12 Taylor in two variables

VII.12 . OPTIMIZATION (8)

- 12.01 Critical points in $\mathbb{R}^2$
- 12.02 Second derivative test (multi)
- 12.03 Absolute extrema on regions
- 12.04 Lagrange multipliers
- 12.05 Constrained optimization apps
- 12.06 Least squares as opt
- 12.07 Gradient descent glimpse
- 12.08 Convexity intro

VII.13 . MULTIPLE INTEGRALS (9)

- 13.01 Double integrals over rectangles
- 13.02 Iterated integrals
- 13.03 Fubini's theorem
- 13.04 Double integrals over regions
- 13.05 Polar double integrals
- 13.06 Applications: mass, center
- 13.07 Triple integrals
- 13.08 Cylindrical coordinates
- 13.09 Spherical coordinates

VII.14 . VECTOR CALCULUS (18)

- 14.01 Vector fields
- 14.02 Line integrals (scalar)
- 14.03 Line integrals (vector)
- 14.04 Conservative fields
- 14.05 Fundamental theorem for lines
- 14.06 Green's theorem
- 14.07 Curl & divergence
- 14.08 Parametric surfaces
- 14.09 Surface integrals (scalar)
- 14.10 Flux integrals
- 14.11 Stokes' theorem
- 14.12 Divergence theorem
- 14.13 Path independence
- 14.14 Potential functions
- 14.15 Applications: work & flow
- 14.16 Maxwell glimpse
- 14.17 Differential forms taste
- 14.18 When theorems apply

VII.15 . FIRST-ORDER EQUATIONS (7)

- 15.01 Review separable & linear
- 15.02 Exact equations
- 15.03 Integrating factors (exact)
- 15.04 Homogeneous first-order
- 15.05 Bernoulli equations
- 15.06 Substitutions
- 15.07 Qualitative analysis

VII.16 . SECOND-ORDER LINEAR (13)

- 16.01 Linear 2nd-order structure
- 16.02 Homogeneous constant coeff
- 16.03 Characteristic equation
- 16.04 Repeated & complex roots
- 16.05 Nonhomogeneous: undetermined
- 16.06 Variation of parameters
- 16.07 Spring-mass systems
- 16.08 Damping regimes
- 16.09 Forced oscillations
- 16.10 Resonance

- 16.11 Cauchy-Euler equations
- 16.12 Boundary value intro
- 16.13 Reduction of order

VII.17 . SYSTEMS OF ODES (11)

- 17.01 First-order systems
- 17.02 Convert higher-order to system
- 17.03 Matrix form $\dot{x}=Ax$
- 17.04 Eigen-method for systems
- 17.05 Phase portraits (2D)
- 17.06 Spiral, node, saddle
- 17.07 Nonhomogeneous systems
- 17.08 Linearization of nonlinear
- 17.09 Predator-prey glimpse
- 17.10 Coupled springs
- 17.11 Stability language

VII.18 . LAPLACE TRANSFORM (9)

- 18.01 Definition & existence
- 18.02 Transforms of basics
- 18.03 s-shift & t-shift
- 18.04 Solve IVP with Laplace
- 18.05 Step & impulse functions
- 18.06 Convolution
- 18.07 Transfer functions glimpse
- 18.08 Partial fractions (Laplace)
- 18.09 Systems via Laplace

VII.19 . FOURIER SERIES (8)

- 19.01 Periodic functions
- 19.02 Fourier coefficients
- 19.03 Sine & cosine series
- 19.04 Convergence idea
- 19.05 Even/odd extensions
- 19.06 Half-range expansions
- 19.07 Parseval glimpse
- 19.08 PDE motivation (heat/wave)